

Multiple integrals – 1

Fubini, the order of integration and symmetry

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Theoretical minimum.

Theorem 1 (Tonelli and Fubini). For $f \geq 0$ measurable on a product the two iterated integrals and the double integral always coincide, finite or not. For general f the same holds as soon as one of the iterated integrals of $|f|$ is finite. Without that hypothesis the order matters: for $f(x, y) = \frac{x^2 - y^2}{(x^2 + y^2)^2}$ on $[0, 1]^2$ the two iterated integrals are $-\frac{\pi}{4}$ and $+\frac{\pi}{4}$.

Theorem 2 (the layer cake formula). For an integrable f on $\mathbb{R}^d$ and $E_\alpha = \{x : |f(x)| > \alpha\}$ one has $\int_{\mathbb{R}^d} |f| = \int_0^\infty m(E_\alpha) d\alpha$, and more generally $\int |f|^p = p \int_0^\infty \alpha^{p-1} m(E_\alpha) d\alpha$. It is Fubini applied to the indicator of the subgraph, and it converts every question about the size of a function into a question about the size of its level sets.

Fact 1 (symmetrisation). If the domain is symmetric under $x \leftrightarrow y$, replacing the integrand by its symmetrisation $\frac{f(x,y)+f(y,x)}{2}$ changes nothing and often trivialises the problem: this is how one gets $\int \int (f(x)-f(y))(g(x)-g(y)) dx dy \geq 0$ for monotone f and g , and with it the integral form of Chebyshev's inequality.

Fact 2 (a double integral as a tool). Writing a one-dimensional quantity as a double integral and then changing the order is one of the cheapest tricks in analysis: $\frac{1}{n} = \int_0^1 x^{n-1} dx$ turns a series into an integral, and $\frac{1}{1-xy} = \sum_{n \geq 0} x^n y^n$ turns an integral into a series. The same passage underlies Cauchy's formula $\int_0^x \int_0^{t_1} \dots \int_0^{t_{n-1}} f = \frac{1}{(n-1)!} \int_0^x (x-t)^{n-1} f(t) dt$.

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1. Prove Fubini's theorem for a continuous function on a rectangle and Tonelli's statement for non-negative functions; then verify that for

$f(x, y) = \frac{x^2 - y^2}{(x^2 + y^2)^2}$ on $[0, 1]^2$ the two iterated integrals differ, and explain which hypothesis fails.

2. (S.T. Yau CSMC 2013) Let f be integrable on $\mathbb{R}^d$ and for $\alpha > 0$ put $E_\alpha = \{x : |f(x)| > \alpha\}$. Prove that $\int_{\mathbb{R}^d} |f(x)| dx = \int_0^\infty m(E_\alpha) d\alpha$, and deduce the corresponding formula for $\int |f|^p$.
3. Prove that $\int_0^1 \int_0^1 \frac{dx dy}{1-xy} = \zeta(2)$ and $\int_0^1 \int_0^1 \frac{dx dy}{1+xy} = \frac{\pi^2}{12}$, justifying the interchange of summation and integration, and find the analogous triple integral representing $\zeta(3)$.
4. (IMS 2009) Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ be non-negative and continuous with $\iint_{\mathbb{R}^2} f(x, y) dx dy = 1$. Prove that there exists a closed disc D of the least possible radius such that $\iint_D f(x, y) dx dy = \frac{1}{2}$.
5. Prove Cauchy's formula for repeated integration, $\int_0^x \int_0^{t_1} \dots \int_0^{t_{n-1}} f(t_n) dt_n \dots dt_1 = \frac{1}{(n-1)!} \int_0^x (x-t)^{n-1} f(t) dt$, by Fubini on the simplex, and use it to compute the volume of the simplex $\{t_1 \geq \dots \geq t_n \geq 0, t_1 \leq 1\}$.